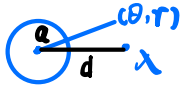
① 分离变量



$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2}{\partial\phi^2}+\frac{\partial^2}{\partial z^2}$$

$$= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}$$

② $\nabla^2\varphi = 0$ $\quad \varphi = R(r)\,\Theta(\theta)$

$\left.\frac{\partial\varphi}{\partial r}\right|_{r=a} = 0$ $\quad \varphi|_{r\to\infty}$ 有界

$\varphi|_{\theta=0} = \varphi|_{\theta=2\pi}$ $\quad \left.\frac{\partial\varphi}{\partial\theta}\right|_{\theta=0} = \left.\frac{\partial\varphi}{\partial\theta}\right|_{\theta=2\pi}$

$$\varphi = (A_0+B_0\ln r)+\sum_{m=1}^{\infty}(A_m r^m+B_m r^{-m})\cos m\theta$$

$$\nabla^2\phi(r,\theta) = -\frac{1}{\varepsilon_0}\rho(r,\theta) = -\frac{I}{6d}\delta(r-d)\delta(\theta)$$

$\rho = C\delta(r-d)\delta(\theta)$ $\quad$ C 6, 电导率 T)

$$\int \rho\, r\,dr\,d\theta = \frac{\varepsilon_0 I}{6} \Rightarrow C = \frac{\varepsilon_0 I}{6d}$$

$$\begin{cases}\nabla^2\phi = -\frac{I}{\sigma d}\delta(r-d)\delta(\theta)\\ \left.\frac{\partial\phi(r,\theta)}{\partial r}\right|_{r=a}=0\end{cases}$$

$$I = \sigma\int d\vec{S}\cdot\vec{E} = -\sigma r\int_0^{2\pi}d\theta\left.\frac{\partial\phi(r,\theta)}{\partial r}\right|_{r\to\infty}$$

$$\phi(r,\theta) = \frac{I}{2\pi\sigma}\ln\sqrt{r^2+d^2-2rd\cos\theta} + \phi'(r,\theta) \quad ②$$

$$\begin{cases}\nabla^2\phi'(r,\theta)=0\\ 0=\frac{\partial}{\partial r}\left[\phi'(r,\theta)+\frac{I}{2\pi\sigma}\ln\sqrt{r^2+d^2-2rd\cos\theta}\right]_{r=a}\\ \left.\frac{\partial\phi'(r,\theta)}{\partial r}\right|_{r\to\infty}\text{ 不含 } r \text{ 正幂次}\end{cases}$$

求出代入①

$$\ln(r^2+r'^2-2rr'\cos\theta)^{\frac{1}{2}} = -\sum_{n=1}^{\infty}\frac{1}{n}\left(\frac{r_<}{r_>}\right)^n\cos n\theta+\ln r_>$$

$$\phi'(r,\theta) = \sum_{m=1}^{\infty} r^m(A_m\cos m\theta+B_m\sin m\theta)+B_0\ln r+A_0$$

$$-\frac{B_0}{a}-\sum_{m=1}^{\infty}m\frac{-A_m\cos m\theta+B_m\sin m\theta}{a^{m+1}} = \frac{I}{2\pi\sigma}\frac{\partial}{\partial r}\left(\ln d-\sum_{m=1}^{\infty}\frac{1}{m}\frac{r^m}{d^m}\cos m\theta\right)$$

$$= -\frac{I}{2\pi\sigma}\left(\sum_{m=1}^{\infty}\frac{r^{m-1}}{d^m}\cos m\theta\right)$$

$B_0 = 0$ $\quad A_m = -\frac{I}{2\pi\sigma}\frac{a^{2m}}{d^m}\frac{1}{m}$

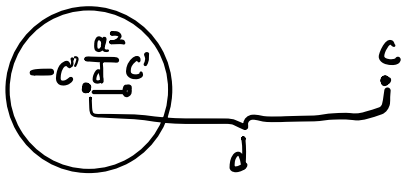
$$\phi'(r,\theta) = \sum_{m=1}^{\infty}-\frac{I}{2\pi\sigma}\frac{(a^2/d)^m}{r^m}\frac{1}{m}\cos m\theta$$

$$= \frac{I}{2\pi\sigma}\left[\ln\left(r^2+\frac{a^4}{d^2}-2r\frac{a^2}{d}\cos\theta\right)^{\frac{1}{2}}-\ln r\right]$$

③ 电像法与 Green 函数

$$\begin{cases}\nabla^2 G(\vec{r},\vec{r}') = -\frac{1}{\varepsilon_0}\delta(\vec{r}-\vec{r}')\\ \left.\frac{\partial G(\vec{r},\vec{r}')}{\partial\vec{r}}\right|_{|\vec{r}|=a}=0\end{cases}$$

由电像法



$$\phi = \frac{1}{2\pi\varepsilon_0}\left[C\ln|\vec{r}-\vec{d}|+C'\ln\left|\vec{r}-\frac{a^2}{d^2}\vec{d}\right|-C''\ln r\right.$$

$$\vec{E} = \frac{1}{2\pi\varepsilon_0}\left[\frac{C(\vec{r}-\vec{d})}{|\vec{r}-\vec{d}|^2}+\frac{C'(\vec{r}-\frac{a^2}{d^2}\vec{d})}{|\vec{r}-\frac{a^2}{d^2}\vec{d}|^2}-\frac{C''\vec{r}}{r^2}\right]$$

$\vec{E}\cdot\frac{\vec{r}}{r} = 0 \Rightarrow C'' = C' = C$

则 $G(\vec{r},\vec{r}')$

$$= \frac{1}{2\pi\varepsilon_0}\left(\ln|\vec{r}-\vec{r}'|+\ln\left|\vec{r}-\frac{a^2}{r'^2}\vec{r}'\right|-\ln|\vec{r}|\right)$$

$G(\vec{r},\vec{r}') = G(\vec{r}',\vec{r})$

于是有 $G(\vec{r}',\vec{r})$ 可解 $\phi(\vec{r})$

$$\phi(\vec{r}) = \int_V d\tau'\rho(\vec{r}')G(\vec{r}',\vec{r})+\varepsilon_0\oint_S ds'\left[G(\vec{r}',\vec{r})\frac{\partial\phi(\vec{r}')}{\partial n'}-\phi(\vec{r}')\frac{\partial G(\vec{r}',\vec{r})}{\partial n'}\right]$$

! 是常数 $\Leftrightarrow$ 与 $\vec{r}$ 无关

当 $\left.\frac{\partial G(\vec{r}',\vec{r})}{\partial n'}\right|_{|\vec{r}'|=a} = \left.\frac{\partial\phi(\vec{r}')}{\partial n'}\right|_{|\vec{r}'|=a} = 0$ 时

$$\phi(\vec{r}) = \int_V d\tau'\rho(\vec{r}')G(\vec{r}',\vec{r})$$

更多：lys 讲义，往年题

$\left(\varphi(\vec{r})\nabla G(\vec{r};\vec{r}') - G(\vec{r};\vec{r}')\nabla\varphi(\vec{r})\right)$的发散项系数是$\vec{r}'$的函数，那么我们就不能进行重整化的操作（发散项对电势有真实物理的无穷大的贡献）。幸运的是，本题中我们并不需要进行重整化操作，因为r很大时，$\varphi = c_1 \ln r + c_2 + O(r^{-1})$，$G(\vec{r};\vec{r}') = c_3 \ln r + c_4 + O(r^{-1})$，则

$$\oiint_{\Sigma_\infty} d\vec{S}\cdot\left(\varphi(\vec{r})\nabla G(\vec{r};\vec{r}') - G(\vec{r};\vec{r}')\nabla\varphi(\vec{r})\right)$$

$$= \lim_{r\to\infty} 2\pi r\left(\left(c_1 \ln r + c_2 + O(r^{-1})\right)\left(\frac{c_3}{r} + O(r^{-2})\right) - \left(c_3 \ln r + c_4 + O(r^{-1})\right)\left(\frac{c_1}{r} + O(r^{-2})\right)\right)$$

$$= 2\pi(c_2 c_3 - c_1 c_4) \qquad (29)$$

无穷远边界对电势的贡献最多为(29)式。如果该式与r'无关，则无穷远边界可以忽略，或者我们先忽略无穷远边界，最后验证(29)式与r'无关是自洽的。

-e 起 e　+v →　← -v

$\vec{v}_- = v\vec{e}_x\,\theta(-t)$　$\vec{a}_- = -v\vec{e}_x\,\delta(t)$

$\vec{v}_+ = -v\vec{e}_x\,\theta(-t)$　$\vec{a}_+ = v\vec{e}_x\,\delta(t)$

$\vec{r}_- = vt\,\theta(-t)$

$\vec{r}_+ = -vt\,\theta(-t)$

考虑 $t^*=0$ 时刻，$\vec{v}^*_\pm = \mp v\vec{e}_x$ 或 $0 = \mp\vec{v}$

$\vec{a}_\pm = \mp v\vec{e}_x\,\delta(t)$

$$\vec{E}_{辐} = \frac{\sum q_\pm}{4\pi\varepsilon_0 c^2 S^{*3}}\left(\vec{R}^*\times\left[\left(c\vec{R}^* - \frac{R^*\vec{v}^*}{c}\right)\times\vec{a}^*\right]\right)$$

$$= \frac{\mu_0 q}{4\pi r^3}\left(\frac{\vec{r}\times\left[\left(\vec{r}+\frac{\vec{v}'}{c}r\right)\times\vec{v}\right]}{\left(1+\frac{\vec{r}\cdot\vec{v}}{rc}\right)^3} + \frac{\vec{r}\times\left[\left(\vec{r}-r\vec{v}/c\right)\times\vec{v}\right]}{\left(1-\frac{\vec{r}\cdot\vec{v}}{rc}\right)^3}\right)\delta(t-r/c)$$

（负电荷　正电荷）

$$= \frac{\mu_0 q}{2\pi r^3}\frac{1+3\left(\frac{\vec{r}\cdot\vec{v}}{rc}\right)^2}{\left[1-\left(\frac{\vec{r}\cdot\vec{v}}{rc}\right)^2\right]^3}\vec{r}\times(\vec{r}\times\vec{v})\,\delta(t-r/c)$$

$$= \frac{\mu_0 q v}{2\pi r}\frac{1+3\beta^2\cos^2\theta}{(1-\beta^2\cos^2\theta)^3}\sin\theta\,\delta(t-\tfrac{r}{c})\,\vec{e}_\theta$$

$$\vec{B}_{辐} = \frac{\vec{R}^*}{cR^*}\times\vec{E}_{辐} = \frac{\mu_0 q v/c}{2\pi r}\frac{1+3\beta^2\cos^2\theta}{(1-\beta^2\cos^2\theta)^3}\sin\theta\,\delta(t-\tfrac{r}{c})\,\vec{e}_\varphi$$

$$\vec{S}_{辐} = \frac{1}{\mu_0 c}E^2_{辐}\,\vec{e}_r$$

wq: $\beta'=0$，则

$$\left.\begin{aligned}\vec{E}_{辐} &= \frac{\mu_0 q v}{2\pi r}\sin\theta\,\delta(t-\tfrac{r}{c})\,\vec{e}_\theta\\ \vec{B}_{辐} &= \frac{\mu_0 q v}{2\pi rc}\sin\theta\,\delta(t-\tfrac{r}{c})\,\vec{e}_\varphi\\ \vec{S}_{辐} &= \frac{\mu_0 q^2 v^2}{4\pi^2 r^2 c}\sin^2\theta\,\delta^2(t-\tfrac{r}{c})\,\vec{e}_r\end{aligned}\right\}\ ①式$$

$$\frac{dI}{d\Omega} = \vec{S}_{辐}r^2 = \frac{\mu_0 q^2 v^2}{4\pi^2 c}\sin^2\theta\,\delta^2(t-\tfrac{r}{c})\,\vec{e}$$

时间反演 $t\to -t$　$\beta'\to-\beta'$　过程：分开

-e 起 e　-v ←　→ v

$\vec{v}_- = -v\vec{e}_x\,\theta(t)$　$\vec{a}_- = -v\vec{e}_x\,\delta(t)$

$\vec{v}_+ = v\vec{e}_x\,\theta(t)$　$\vec{a}_+ = v\vec{e}_x\,\delta(t)$

$\vec{r}_- = -vt\,\theta(t)$

$\vec{r}_+ = vt\,\theta(t)$

$$\vec{E}^{(推)}_{辐}\ \underset{\beta'\to-\beta'}{t\to-t} = \frac{\mu_0 q v}{2\pi r}\frac{1+3\beta^2\cos^2\theta}{(1-\beta^2\cos^2\theta)^3}\sin\theta\,\delta(t+\tfrac{r}{c})\,\vec{e}_\theta = \frac{\mu_0 q v}{2\pi r}\sin\theta\,\delta(t+\tfrac{r}{c})\,\vec{e}_\theta$$

$\vec{B}^{(推)}_{辐} = \frac{\vec{R}^*}{cR^*}\times\vec{E}_{辐}$.　$\vec{S}_{辐}$　$\frac{dI}{d\Omega}$　略

} ②

☆ $\delta(-t) = \delta(t)$

直接用推广写？

$\delta(t-\frac{r}{c})$

↑

$$(r, r/c^{(-)}):\begin{pmatrix}\varphi\\ \vec{A}\end{pmatrix} = \frac{q}{4\pi\varepsilon_0 r}\frac{2}{1-\beta^2\cos^2\theta}\begin{pmatrix}-\beta\cos\theta\\ \vec{\beta}/c\end{pmatrix}$$

$$(r, r/c^{(+)})\begin{pmatrix}\varphi\\ \vec{A}\end{pmatrix} = 0$$

$$t^* = t - \frac{R^*}{c}$$

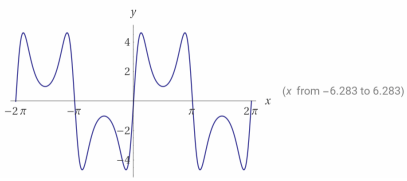
$$S^* = R^*\left(1-\frac{\vec{R}^*\cdot\vec{v}^*}{R^*c}\right)\quad \vec{R}^* = \vec{r}-\vec{r}^* = \vec{r}$$

$\vec{r} = (x, y, z) = (r, \theta)$ 中心对称

Result

$$\frac{\sin(x)\left(\frac{3\cos^2(x)}{2}+1\right)}{\left(1-\frac{\cos^2(x)}{2}\right)^3}$$

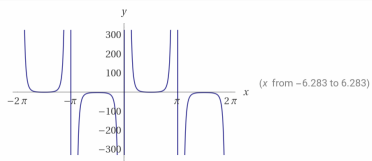
Plots



Result

$$\frac{\sin(x)\left(3\cos^2(x)+1\right)}{\left(1-\cos^2(x)\right)^3}$$

Plots



超前势写碰撞 ⊡式 $C \to -C$

$\vec{E}_{辐}^{(超)} = \vec{E}_{辐}^{(推)}$ $\vec{B}_{辐}^{(推)} = -\vec{B}_{辐}^{(超)}$

格林函数

$$\begin{cases}\nabla^2\phi(\vec{r})=-\frac{1}{\varepsilon_0}\rho(\vec{r}) \\ \text{边条件}\end{cases}$$

分离变量 → $\nabla^2\phi=0$，$\phi=\int\frac{\rho d\tau'}{4\pi\varepsilon_0 R}$，边条件

格林函数 → $\nabla'^2 G(\vec{r},\vec{r}')=-\frac{1}{\varepsilon_0}\delta$，边条件.

$$\phi(\vec{r})=\int d\tau'\,\phi(\vec{r}')\,\delta(\vec{r}'-\vec{r})$$

$$=-\varepsilon_0\int_\tau d\tau'\,\phi(\vec{r}')\,\nabla'^2 G(\vec{r}',\vec{r})$$

$$=\varepsilon_0\int_\tau d\tau'\left\{\nabla'\cdot[-\phi(\vec{r}')\nabla' G(\vec{r}',\vec{r})+[\nabla'\phi(\vec{r}')]G(\vec{r}',\vec{r})-\nabla'^2\phi(\vec{r}')G(\vec{r}',\vec{r})\right\}$$

$$=\int_\tau d\tau'\rho(\vec{r}')G(\vec{r}',\vec{r})+\varepsilon_0\oint_S dS'\left[-G(\vec{r}',\vec{r})\frac{\partial\phi(\vec{r}')}{\partial n'}-\phi(\vec{r}')\frac{\partial G(\vec{r}',\vec{r})}{\partial n'}\right]$$

第一类边界 ϕ 已知，$G(\vec{r}',\vec{r})|_S=0$

$$\phi(\vec{r})=\int_\tau d\tau'\rho(\vec{r}')G(\vec{r}',\vec{r})-\varepsilon_0\oint_S dS'\phi(\vec{r}')\frac{\partial G(\vec{r}',\vec{r})}{\partial n'}$$

第二类 $\frac{\partial\phi}{\partial n'}|_S$ 已知，$\frac{\partial G(\vec{r}',\vec{r})}{\partial n'}|_S=C$ $(=-\varepsilon_0 C)$

$\phi(\vec{r}')$ 积分是与 $\vec{r}'$ 无关常数!

$$\phi(\vec{r})=\int_\tau d\tau'\rho(\vec{r}')G(\vec{r}',\vec{r})+\varepsilon_0\oint_S dS' G(\vec{r}',\vec{r})\frac{\partial\phi(\vec{r}')}{\partial n'}$$

注 $1=\int_\tau d\tau'\delta(\vec{r}'-\vec{r})=-\varepsilon_0\int_\tau d\tau'\nabla'^2 G(\vec{r}',\vec{r})$

$$=-\varepsilon_0\oint_S dS'\frac{\partial G(\vec{r}',\vec{r})}{\partial n'}$$

无界格林函数

$$G(\vec{r},\vec{r}')=\frac{1}{4\pi\varepsilon_0}\frac{1}{|\vec{r}-\vec{r}'|}$$

上半空间 $z>0$，第一类边值

$$G(\vec{r},\vec{r}')=\frac{1}{4\pi\varepsilon_0}\frac{1}{|\vec{r}-\vec{r}'|}-\frac{1}{|\vec{r}-\vec{x}-\vec{y}+\vec{z}'|}$$

球外空间?

$$G(\vec{r},\vec{r}')=\frac{1}{4\pi\varepsilon_0}\left(\frac{1}{|\vec{r}-\vec{r}'|}-\frac{\frac{r_0}{r'}}{|\vec{r}-\frac{r_0^2}{r'^2}\vec{r}'|}\right)$$

推迟势

$$\begin{cases}\vec{E}=-\nabla\phi-\frac{\partial\vec{A}}{\partial t} \\ \vec{B}=\nabla\times\vec{A}\end{cases}$$

$$\begin{cases}\phi(\vec{r},t)=\frac{q}{4\pi\varepsilon_0 R^*}\frac{1}{1-\frac{\vec{R}^*\cdot\vec{v}^*}{R^* c}} \\ \vec{A}(\vec{r},t)=\frac{\mu_0 q\vec{v}^*}{4\pi R^*}\frac{1}{1-\frac{\vec{R}^*\cdot\vec{v}^*}{R^* c}}\end{cases}$$

$$\begin{cases}\vec{R}^*=\vec{r}-\vec{r}^* \\ \vec{r}^*=\vec{r}^*(\vec{r},t)=\vec{r}^*(t^*)\end{cases}$$

$$\frac{\partial t^*}{\partial t}=\frac{1}{1-\frac{\vec{R}^*\cdot\vec{v}}{R^* c}}\qquad\frac{\partial R^*}{\partial t}=\frac{-\frac{\vec{R}^*}{R}\cdot\vec{v}^*}{1-\frac{\vec{R}^*\cdot\vec{v}}{R^* c}}$$

$$\nabla t^*=\frac{-\vec{R}^*/cR^*}{1-\frac{\vec{R}^*\cdot\vec{v}}{R^* c}}\qquad\nabla R^*=\frac{\vec{R}^*/R}{1-\frac{\vec{R}^*\cdot\vec{v}^*}{R^* c}}$$

$$\frac{\partial\vec{R}^*}{\partial t}=-\vec{v}^*\frac{\partial t^*}{\partial t}\qquad\nabla\vec{R}^*=\vec{I}-\nabla t^*\cdot\vec{v}^*$$

$$\vec{E}(\vec{r},t)=\frac{q}{4\pi\varepsilon_0 S^{*3}}\left(\vec{R}^*-\frac{R^*\vec{v}^*}{c}\right)\left(1-\frac{v^{*2}}{c^2}\right)+\frac{q}{4\pi\varepsilon_0 c^2 S^{*3}}\left\{\vec{R}^*\times\left[\left(\vec{R}^*-\frac{R^*\vec{v}^*}{c}\right)\times\vec{a}^*\right]\right\}$$

$$\vec{B}=\frac{\vec{R}^*}{cR^*}\times\vec{E}\qquad\left(S^*=R^*\left(1-\frac{\vec{R}^*\cdot\vec{v}^*}{R^* c}\right)\right)$$

$$\vec{S}=\frac{1}{\mu_0}(\vec{E}_{辐}\times\vec{B}_{辐})$$

$$=\frac{1}{\mu_0 c}(E_{辐}^2\vec{n}^*-(\vec{n}^*\cdot\vec{E}_{辐})\vec{E}_{辐})\quad(\vec{n}^*\perp\vec{E}_{辐}\Rightarrow\vec{n}^*\cdot\vec{E}_{辐}=0)$$

$$=\frac{q^2\vec{n}^*}{16\pi^2\varepsilon_0 c^5 S_0^{*6}\mu_0}\frac{|\vec{n}\times[(\vec{n}^*-\frac{\vec{v}^*}{c})\times\vec{a}^*]|^2}{(1-\frac{\vec{n}^*\cdot\vec{v}^*}{c})^6}$$

